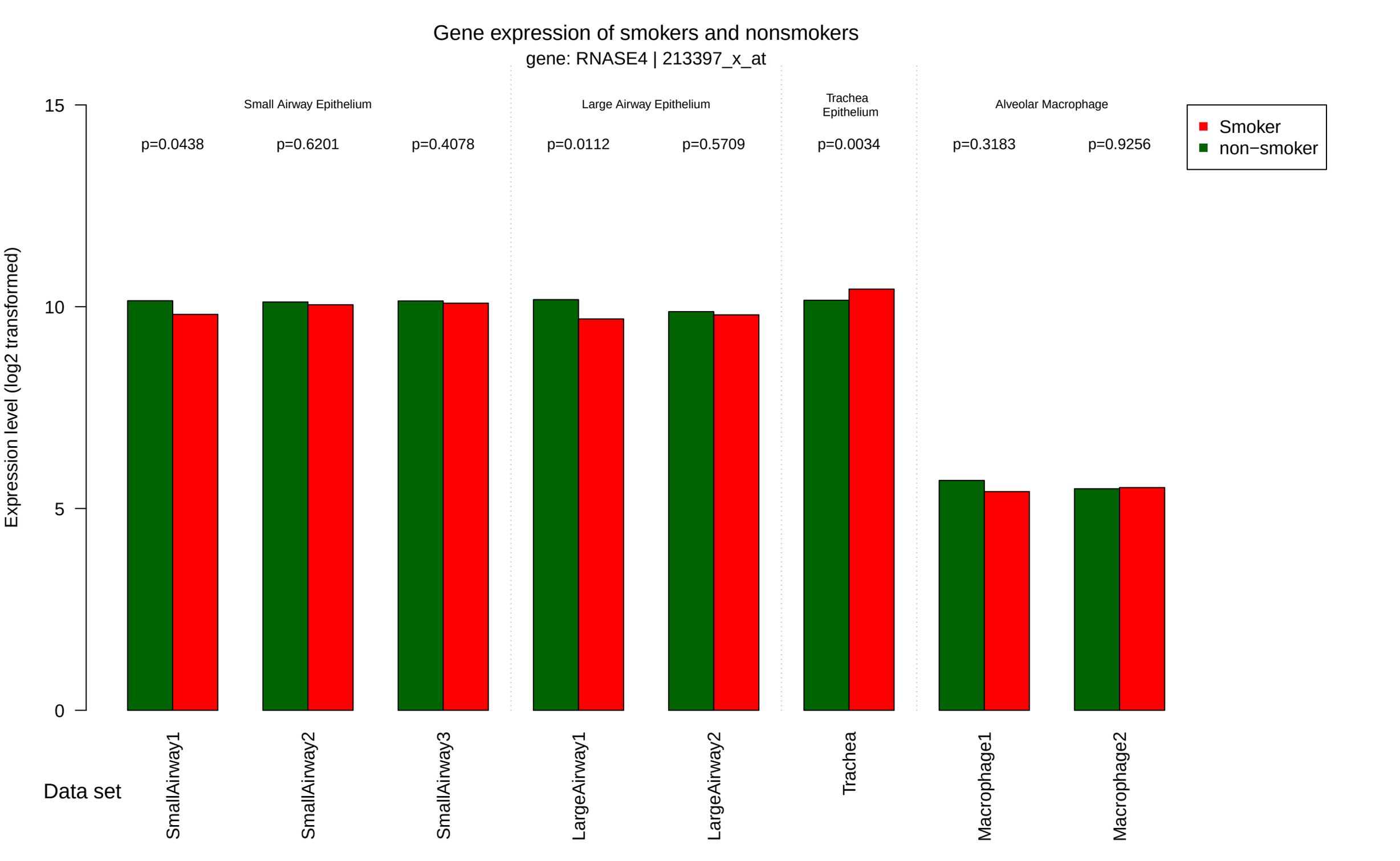
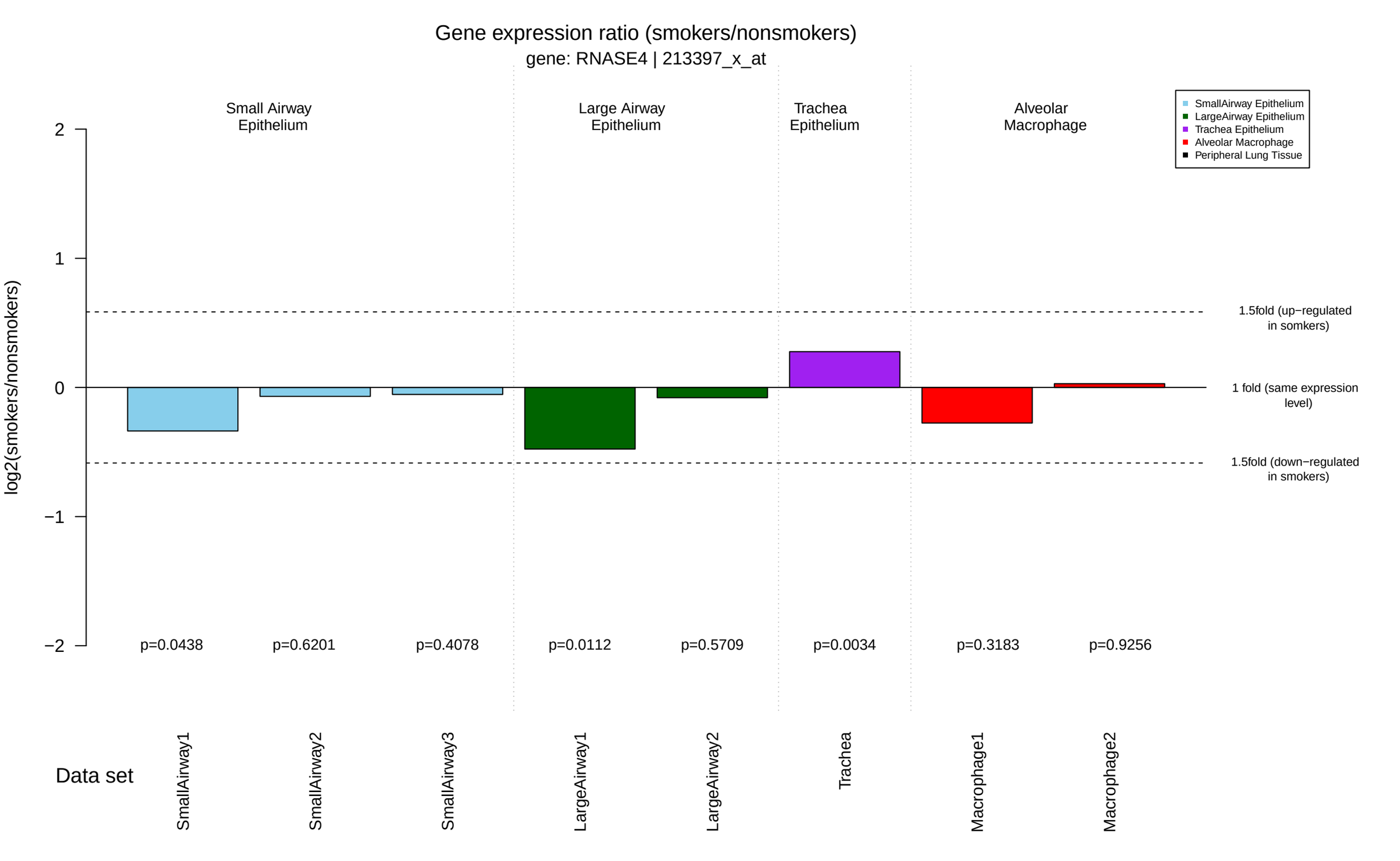
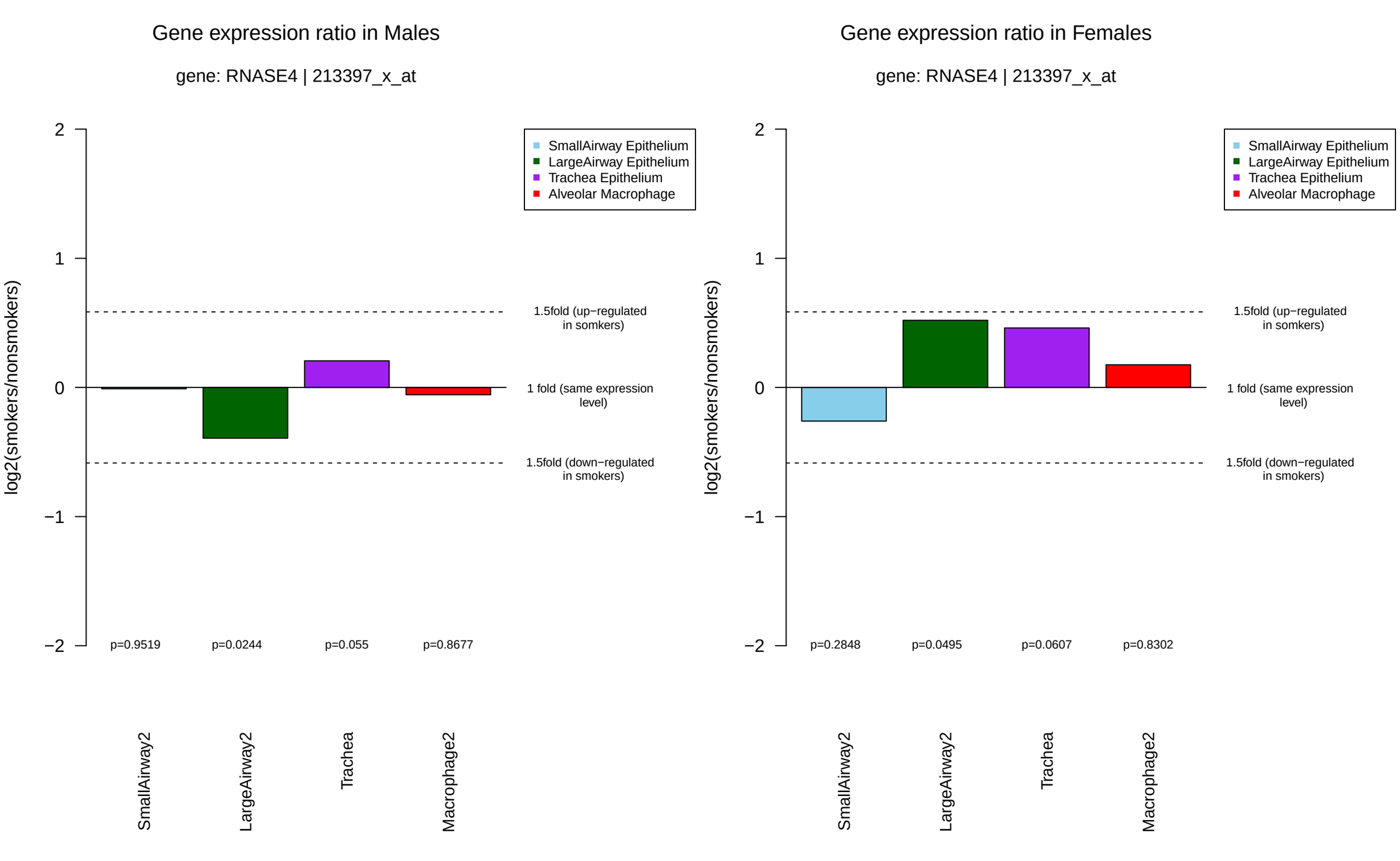
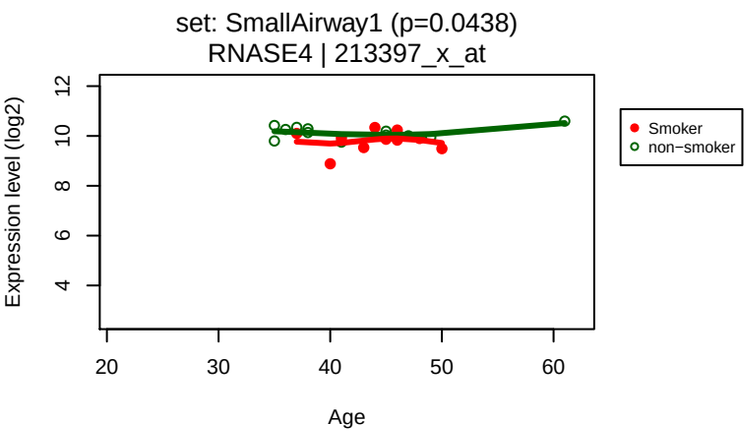
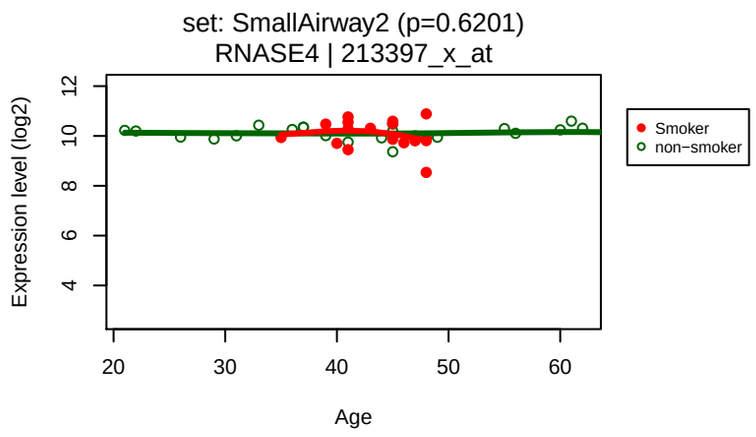
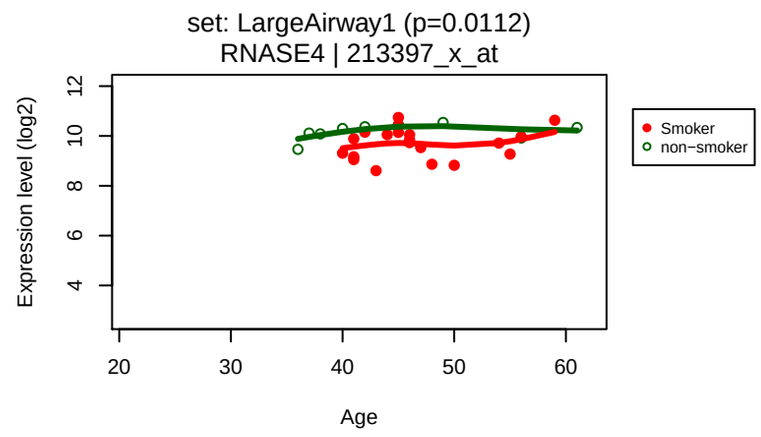

Gene expression of smokers and nonsmokers
gene: RNASE4 | 213397_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0438
p=0.6201
p=0.4078
p=0.0112
p=0.5709
p=0.0034
p=0.3183
p=0.9256
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

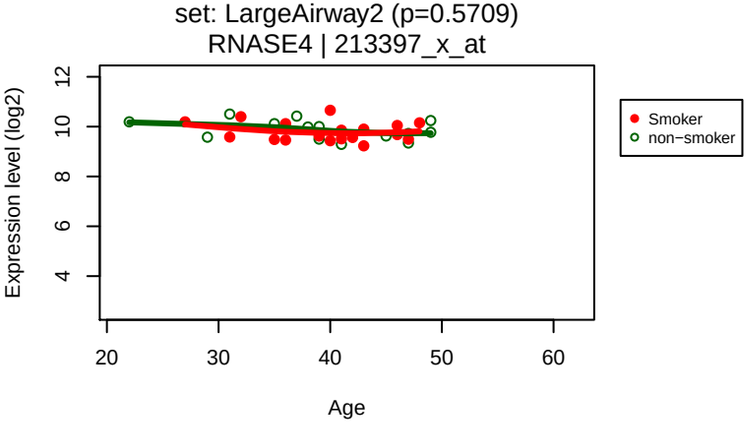

Gene expression ratio (smokers/nonsmokers)
gene: RNASE4 | 213397_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0438
p=0.6201
p=0.4078
p=0.0112
p=0.5709
p=0.0034
p=0.3183
p=0.9256
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

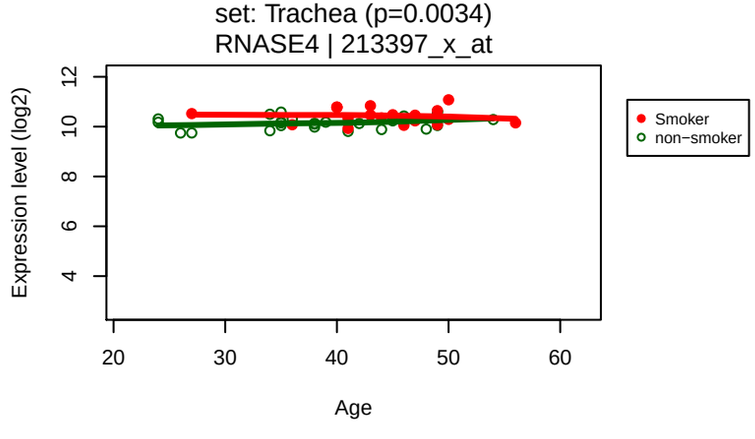

Gene expression ratio in Males
gene: RNASE4 | 213397_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9519
p=0.0244
p=0.055
p=0.8677
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNASE4 | 213397_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2848
p=0.0495
p=0.0607
p=0.8302
SmallAirway2
LargeAirway2
Trachea
Macrophage2

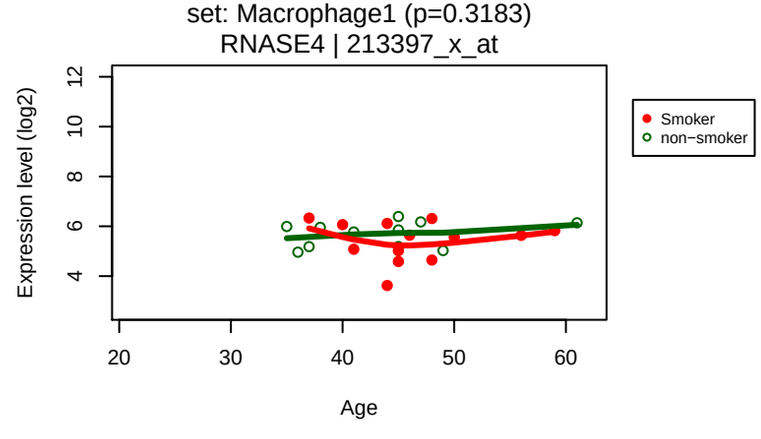

set: SmallAirway1 (p=0.0438)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker

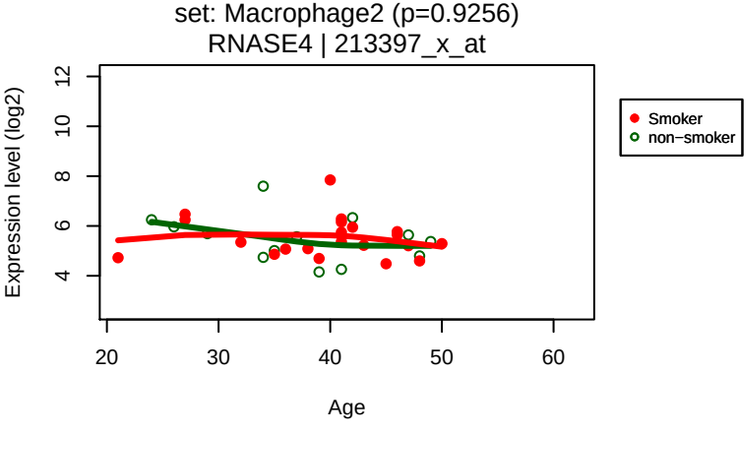

set: SmallAirway2 (p=0.6201)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker

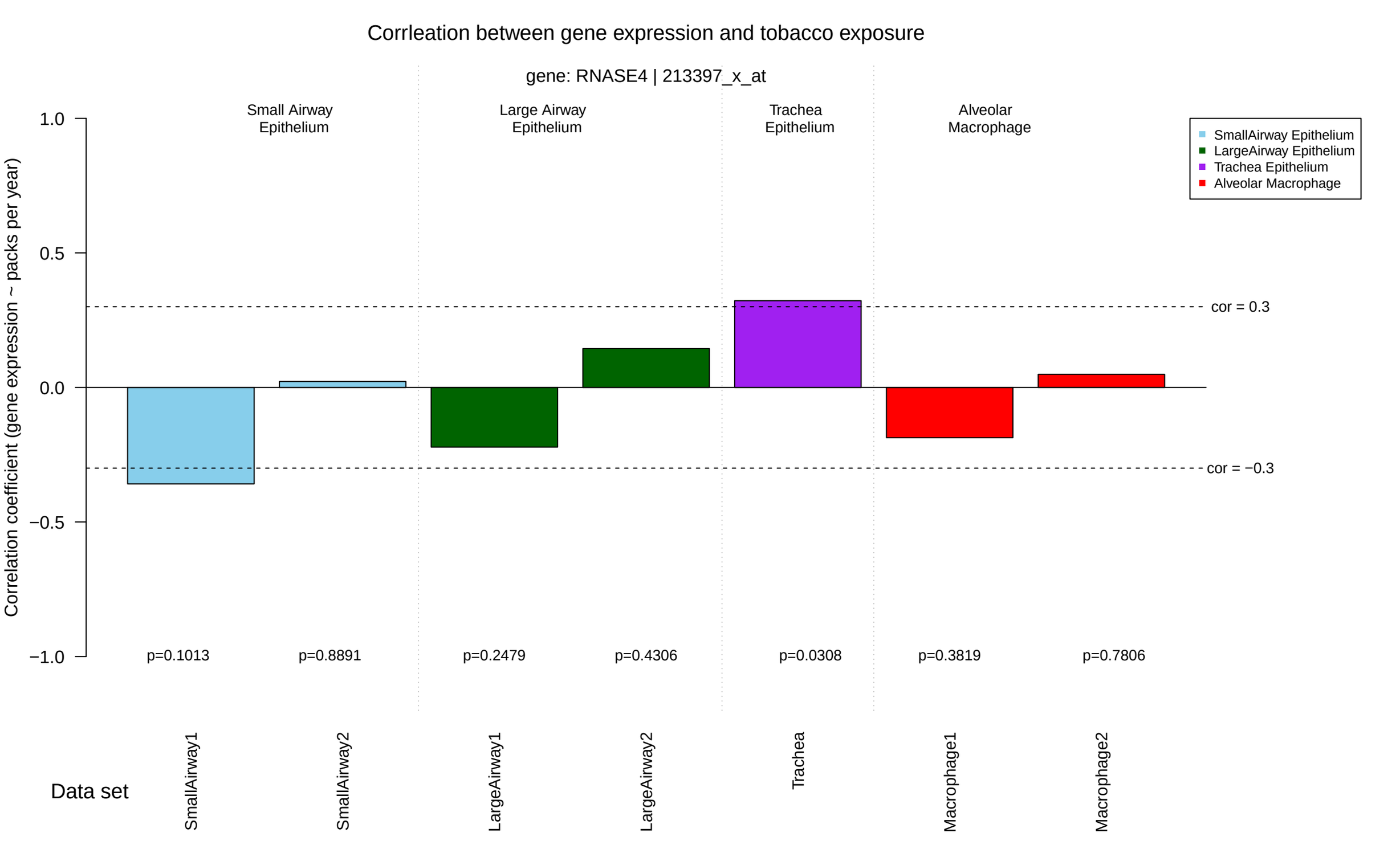

set: LargeAirway1 (p=0.0112)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5709)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0034)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3183)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9256)
RNASE4 | 213397_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNASE4 | 213397_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1013
p=0.8891
p=0.2479
p=0.4306
p=0.0308
p=0.3819
p=0.7806
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2